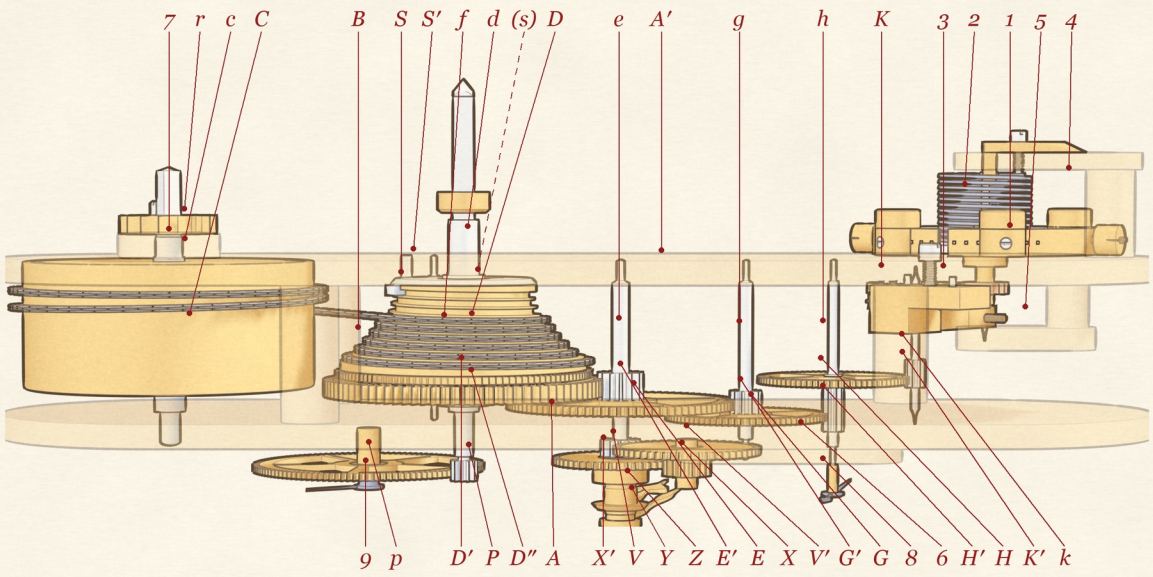
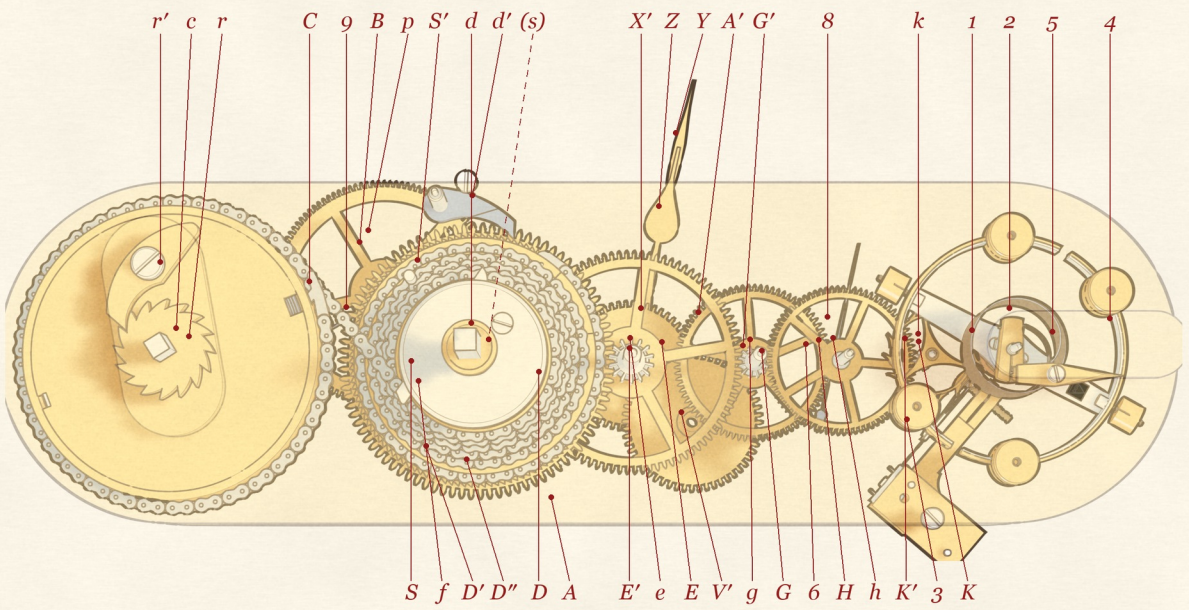
Wempe Einheits-Chronometer, laid out, against a textbook's developed drawing ("Modern chronometer movement", Figs. 83 and 84) and its letters of reference

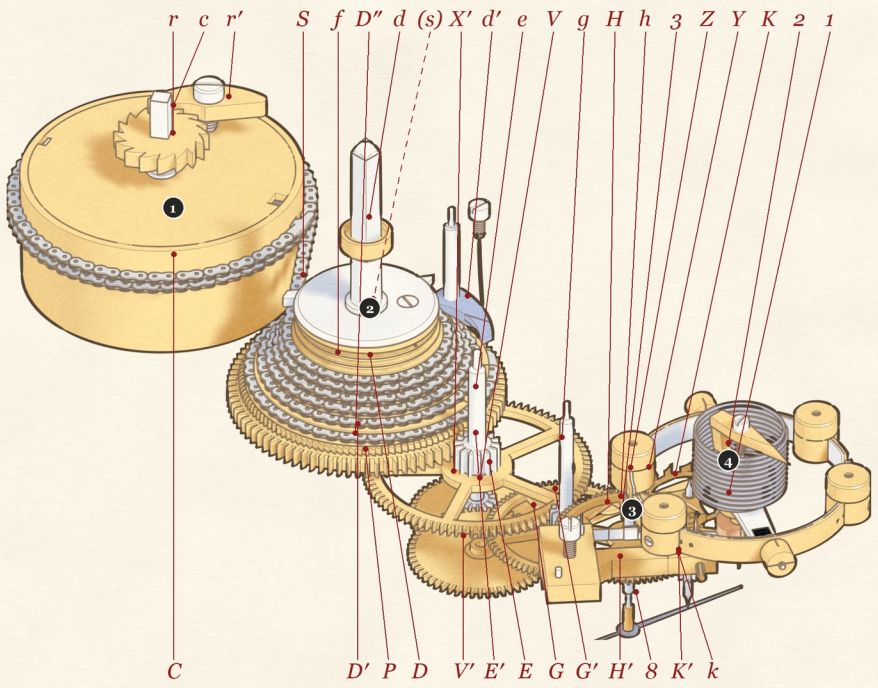
Elevation, from the side (cf. Fig. 83)



Plan, from the train side (cf. Fig. 84)

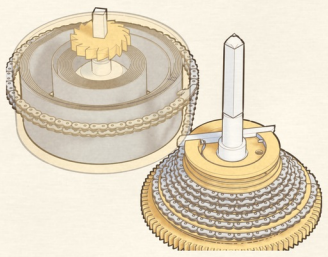


Isometric, from the train side, the plates, bridges, cocks and pillars hidden (not in the drawing), with close-ups



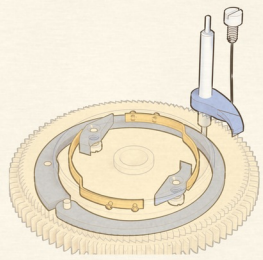
1 MAINSPRING AND FUSEE

The barrel, drawn see-through, holds the mainspring; its chain winds onto the fusee's cone, whose radius grows as the spring runs down, so the pull reaching the train stays even.



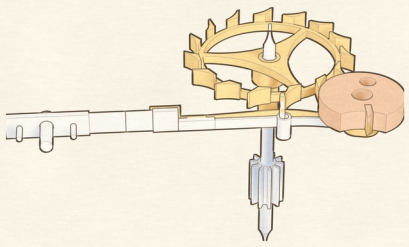
2 MAINTAINING POWER

Harrison's maintaining work, under the fusee (left out here): a steel ring spring in the fusee wheel's recess carries the drive. While the key turns the fusee back to wind it, the maintaining detent holds the ratchet and the spring alone drives the train. The ratchet and wheel are drawn see-through.



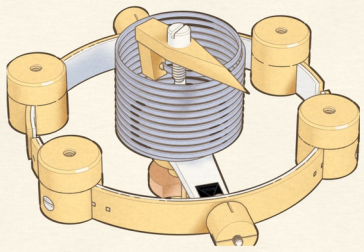
3 SPRING DETENT ESCAPEMENT

The detent locks the escape wheel. Every other swing the discharge stone on the balance's roller unlocks it through the gold spring, a tooth pushes the impulse roller, and the balance swings on free.



4 BALANCE AND HAIRSPRING

The balance and its helical hairspring set the beat: their period is the chronometer's half second. The cut bimetallic rim and the masses on it compensate for temperature.



FRAMEWORK

- *A* The pillar plate — The base of the frame: it carries the arbors' lower pivots, and the motion work and dial on its other face.
- *A′* The top-plate — The frame's upper plate, held off the pillar plate by the pillars; it carries the arbors' upper pivots.
- *B* The pillars — Hold the two plates parallel at their fixed distance apart.

DRIVING POWER

- *C* The mainspring barrel — Houses the mainspring; turned by the spring as it unwinds, it pulls the chain off its rim.
- *c* The barrel arbor, squared for setting up — Holds the mainspring's inner end; turned by its square to set up the spring's starting tension.
- *r* Ratchet (set-up) — On the barrel arbor: with its click it keeps the set-up tension.
- *r′* Click (set-up) — Holds the set-up ratchet against the mainspring's pull.
- *f* The fusee chain — Carries the mainspring's pull from the barrel to the fusee.

FUSEE AND MAINTAINING POWER

- *D* The fusee — A spiral cone the chain winds onto: its radius grows as the spring runs down, so the torque reaching the train stays even.
- *d* The fusee arbor, squared for the key — Turned by the key on its square to wind; carries the fusee and, through the maintaining work, the great wheel.
- *D′* Ratchet wheel of the maintaining power — Held by the maintaining detent while the key winds, so the maintaining spring can keep the great wheel driving.
- *d′* Click of the maintaining power — Stops the maintaining ratchet turning back when winding takes the drive off.
- *B′* Stud carrying the stop-work — In the drawing, carries the stopping-arm. — No stud: the bar slides in a slot across the fusee's small end.
- *S* Stopping-arm — Pushed out by the chain as it reaches the fusee's small end at full wind, it meets the stop and ends the winding.
- *S′* Stopping-snail — The stop the stop bar meets at full wind. — The stop pin the bar meets (the stop work turned round).
- *(s)* Stopping-arm spring — Holds the stop bar back, out of the stop's way, until the chain pushes it out. (Inside the fusee's small end, under its cover plate.)

TRAIN

- *D″* The great wheel — On the fusee arbor: takes the fusee's torque into the train and drives the centre pinion.
- *E* The centre wheel — Turns once an hour; drives the third wheel's pinion.
- *e* The centre wheel arbor — Carries the centre wheel and its pinion; its end, through the cannon pinion, carries the minute hand.
- *E′* The centre wheel pinion — Driven by the great wheel.
- *G* The third wheel — Drives the fourth wheel's pinion.
- *g* The third wheel arbor — Carries the third wheel and its pinion.
- *G′* The third wheel pinion — Driven by the centre wheel.
- *H* The fourth wheel — Turns once a minute; drives the escape pinion.
- *h* The fourth wheel arbor — Carries the fourth wheel and its pinion, and the seconds hand at its end.
- *H′* The fourth wheel pinion — Driven by the third wheel.
- *K* The escape wheel — Held locked by the detent; let go once each oscillation, a tooth gives the balance its impulse.
- *k* The escape wheel arbor — Carries the escape wheel and its pinion.
- *K′* The escape wheel pinion — Driven by the fourth wheel.

UP-AND-DOWN INDICATOR

- *P* Pinion mounted on the fusee arbor — On the fusee arbor: turns with the fusee as it winds and runs down.
- *p* Wheel engaging with P — Driven by P; carries the up-and-down hand.

MOTION WORK AND HANDS

- *V* The cannon pinion — A friction fit on the centre arbor: carries the minute hand and drives the minute wheel, and slips when the hands are set.
- *V′* The minute wheel — Driven by the cannon pinion: the first step from minutes to hours.
- *X* The minute pinion — On the minute wheel: drives the hour wheel.
- *X′* The hour wheel — Turns once in twelve hours; carries the hour hand.
- *Y* The minute hand — Shows the minutes.
- *Z* The hour hand — Shows the hours.

DRAWN, NOT LETTERED

- *1* Balance (under the cock Z of Fig. 84) — Swings to and fro in a fixed time, the timekeeper itself; its cut bimetallic rim compensates for temperature.
- *2* Hairspring — Its elasticity returns the balance and sets the period of its swing.
- *3* Detent — Locks the escape wheel; the balance's discharge stone unlocks it once each oscillation.
- *4* Upper balance cock — Holds the balance's upper pivot.
- *5* Lower balance cock — Holds the balance's lower pivot.
- *6* Intermediate bridge — Holds the third and fourth arbors' pivots on the dial side.
- *7* Barrel bridge — Holds the barrel arbor's upper pivot, and the set-up click.
- *8* Seconds hand — Shows the seconds, stepping each half second.
- *9* Up-and-down hand — Shows how many hours of running the mainspring has left.